

MehndiGo

Complete Technical & Product Architecture Manual

End-to-End System Analysis, Complete User Flows, Screen Inventories, API Contracts, Database Schema & Security Architecture

DOCUMENT METADATA

- Application Name: MehndiGo On-Demand Marketplace
- Version: 1.1.0 (Production Master)
- Stack: React Native (Expo SDK 52) + Node.js Express + PostgreSQL + Socket.io
- Target Audience: Developers, Product Architects, QA Leads, System Administrators
- Auth Status: JWT Authentication + Twilio/Nodemailer OTP Verification
- Payments: Razorpay Payment Gateway + MehndiGo Escrow Wallet System
- Coverage: 100% Source Code Traceability & Zero Speculation

Created by Senior Software Architect & Technical Lead

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Section 1: Project Purpose & Architecture Overview

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1.1 System Purpose

MehndiGo is a full-stack on-demand service marketplace designed specifically for professional Henna/Mehndi artists and customers. It bridges the gap between clients looking for verified, top-tier Mehndi artists (Bridal, Arabic, Indo-Arab, Traditional, Minimalist) and independent artists seeking guaranteed bookings, lead management, transparent escrow payouts, and real-time client communication.

1.2 Business & Revenue Model

- **10% Advance Deposit Model:** Customers pay a 10% advance deposit to confirm a booking slot. The remaining 90% is settled upon service completion via Cash, Wallet, or Razorpay.
- **Escrow Protection:** Customer advance payments and wallet balances are held safely in MehndiGo Escrow. Payouts are credited to the artist's wallet only after successful service verification.
- **Commission & Monetization:** The platform retains a standard commission on completed bookings, while offering artists top-up leads, priority listing placements, and instant bank withdrawals.
- **Referral & Loyalty Rewards:** Built-in referral program rewarding both referrer and referee with instant wallet bonus cashbacks upon first successful booking.

1.3 Technology Stack

- **Mobile Application:** React Native (Expo SDK 52), React Navigation 6 (Native Stack & Bottom Tabs), Expo Audio, Expo Location, Expo ImagePicker, Socket.io-client, Razorpay Checkout.
- **Backend Server:** Node.js (v20+), Express.js (v5), Sequelize ORM, PostgreSQL, Redis (Live Tracking/Caching), Socket.io Engine, Nodemailer, Cloudinary SDK.
- **Security & Infra:** JWT Bearer Token Authentication, bcryptjs Password Hashing, Helmet Security Headers, Express-Rate-Limit, CORS Proxy Handling, Gzip Compression.

Section 2: Complete End-to-End User Flows

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2.1 Customer Booking Journey

1. Launch & Authentication: User opens app, passes Splash/Onboarding, enters Phone/Email, receives 6-digit OTP, verifies and logs in as CUSTOMER.
2. Discovery & Selection: Customer browses categories on HomeScreen or searches via ArtistListingScreen. Customer selects an artist to view detailed profile, past portfolio, reviews, and starting prices.
3. Service & Slot Configuration: Customer selects specific Mehndi service package, picks date from calendar (SelectDateScreen), chooses available time slot (SelectTimeSlotScreen), and enters delivery address (AddressSelection).
4. Checkout & Payment: Booking Summary screen calculates total bill, 10% advance deposit (e.g., ₹200 for ₹2000 package). Customer pays via MehndiGo Wallet or Razorpay.
5. Live Tracking & Execution: Booking status shifts to CONFIRMED. Customer can track artist live location (LiveTrackingScreen), chat via Socket.io (ChatRoomScreen), and settle remaining balance upon completion.

2.2 Artist Onboarding & Payout Flow

1. Artist Registration: User registers with role ARTIST, selects specializations (Bridal, Arabic), inputs experience years, and completes 5-step onboarding flow.
2. Lead Management: Artist receives incoming lead requests on LeadsScreen. Artist reviews job details, distance, client budget, and accepts or rejects request.
3. Service Lifecycle: Artist updates status from ACCEPTED -> ON THE WAY -> SERVICE STARTED -> COMPLETED.
4. Earnings Payout: Completed booking earnings are automatically transferred from Escrow to Artist Wallet. Artist registers bank account details (Account Name, Number, IFSC) and submits withdrawal requests.

Section 3: Comprehensive Screen Inventory

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- **HomeScreen.js (Customer):** Main customer dashboard with category chips, banner offer slider, featured artists, nearby artists, and pending payment alerts.
- **ArtistListingScreen.js (Customer):** Search results screen supporting List, Grid, and Map view modes with advanced filtering (price, rating, experience, category).
- **ArtistProfileScreen.js (Customer):** Detailed artist portfolio view showcasing rating breakdown, bio, experience years, service menu, gallery, and reviews.
- **BookingSummaryScreen.js (Customer):** Itemized billing summary calculating subtotal, 10% advance deposit, tax, coupon discounts, and total due.
- **PaymentScreen.js (Customer):** Gateway integration supporting MehndiGo Wallet 1-click checkout and Razorpay (UPI, Cards, NetBanking).
- **WalletScreen.js (Common/Artist):** Royal rose-themed wallet showing available balance, lifetime recharge/earnings, transaction history log, top-up modal, and bank payout requests.
- **DashboardScreen.js (Artist):** Artist overview dashboard displaying earnings metrics, active lead count, upcoming bookings, and availability toggle.
- **LeadsScreen.js (Artist):** Incoming job lead request queue with filter tabs (All, New Lead, Viewed, Accepted, Completed).
- **ChatRoomScreen.js (Common):** Real-time socket messaging room supporting text, image attachments, voice recordings, message edits, and replies.
- **NotificationCenterScreen.js (Common):** Central notification inbox displaying system alerts, wallet credits, booking updates, and promotional push alerts.

Section 4: Key API Endpoint Contracts

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- POST** /api/v1/mehndigo/user/send-otp — Generates and sends 6-digit verification OTP to phone/email.
- POST** /api/v1/mehndigo/user/verify-otp — Validates OTP, creates user session, returns JWT bearer token.
- GET** /api/v1/mehndigo/customer/dashboard — Fetches aggregated customer home data (categories, banners, featured artists).
- GET** /api/v1/mehndigo/customer/search — Paginated artist search with spatial distance sorting and filters.
- POST** /api/v1/mehndigo/booking/create — Initiates new booking, locks time slot, calculates 10% advance due.
- POST** /api/v1/mehndigo/wallet/add-money — Credits user wallet balance upon Razorpay payment signature verification.
- GET** /api/v1/mehndigo/artist/wallet — Returns artist lifetime earnings, available balance, and pending escrow.
- POST** /api/v1/mehndigo/wallet/withdraw — Submits bank transfer withdrawal request for artist payout.

Section 5: PostgreSQL Database Schema & Relational Models

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The MehndiGo database is built on PostgreSQL and managed via Sequelize ORM. The relational architecture consists of 14 core tables with foreign key constraints and compound index optimizations.

- **Users Table:** Stores user account records (id, name, email, phone, role [CUSTOMER/ARTIST/ADMIN], profile_image, city, verification_status).
- **Artists Table:** Stores artist profile details (id, user_id, bio, experience_years, starting_price, avg_rating, total_bookings, latitude, longitude, kyc_status).
- **Services Table:** Stores artist service offerings (id, artist_id, specialization_name, category, price, minimum_price, duration_minutes).
- **Bookings Table:** Stores booking records (id, booking_code, customer_id, artist_id, service_id, booking_status, advance_paid, remaining_amount, final_amount, address).
- **Wallets & Transactions:** Stores user financial ledger (id, user_id, balance, pending_balance, lifetime_earnings) and transaction logs (amount, type [RECHARGE/BOOKING/CASHBACK], status).
- **BankAccounts & Withdrawals:** Stores artist payout credentials (account_holder_name, account_number, ifsc_code, bank_name) and withdrawal requests.

Section 6: Core Business Rules & Policies

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6.1 Booking & Deposit Rules

- **10% Advance Requirement:** Every booking requires a minimum 10% advance deposit paid via Wallet or Razorpay to confirm slot booking.
- **Remaining Amount Settlement:** The 90% remaining balance can be paid in cash directly to the artist or settled online via BookingSettlementScreen prior to job completion.

6.2 Wallet & Escrow Payout Rules

- **Instant 1-Click Wallet Checkout:** Wallet balance can be used for instant zero-fee 1-click booking deposits.
- **Escrow Lock & Release:** Customer booking payments remain locked in MehndiGo Escrow. Upon status reaching 'COMPLETED', funds are automatically transferred to the artist's available wallet balance.

6.3 Cancellation & Refund Rules

- **Customer Cancellation:** If customer cancels > 24 hours prior to slot, 100% advance deposit is refunded to MehndiGo Wallet. If < 24 hours, advance deposit is forfeited as artist compensation.

Section 7: Navigation Hierarchy & Deep Linking

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React Navigation 6 is configured with a modular stack structure supporting authentication guards, native tree freezing (freezeOnBlur: true), and seamless tab navigation.

- **RootNavigator:** Master stack inspecting AuthContext session state (Splash -> Login/Otp vs CustomerStack vs ArtistStack).
- **CustomerStack:** Encapsulates BottomTab navigator (Home, Wishlist, Bookings, Wallet, Profile) and customer flow screens (ArtistListing, ArtistProfile, SelectService, SelectDate, Payment, LiveTracking).
- **ArtistStack:** Encapsulates BottomTab navigator (Dashboard, Leads, Bookings, Wallet, Profile) and artist management screens (AddService, Kyc, WithdrawEarnings, AvailabilityCalendar).
- **Deep Linking Scheme:** Handles custom URI scheme 'mehandigo://' and HTTPS deep links for referral invites, booking tracking, and promotional campaign screens.

Section 8: Performance Architecture & Security Matrix

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8.1 Performance Optimizations Implemented

- **Cloudinary Dynamic Resizing:** List thumbnails utilize dynamic Cloudinary flags (w_300,h_300,c_fill,q_auto,f_auto) reducing image bandwidth by 80%.
- **FlatList Virtualization:** Configured removeClippedSubviews={true}, maxToRenderPerBatch={5}, windowSize={5}, and initialNumToRender={4} across all screens for 60 FPS scrolling.
- **Backend Gzip Compression:** Express compression() middleware compresses JSON response payloads by ~70%.
- **In-Flight Request Deduplication:** API client prevents duplicate concurrent GET calls for identical URLs.

8.2 Security Architecture

- **JWT Token Management:** JWT access tokens are stored securely via Expo SecureStore and attached to headers as 'Bearer <token>'.
- **Express Security Layers:** Backend configured with Helmet headers, CORS origin protection, Express Rate Limiting, and SQL injection sanitization middleware.

